SAPIENZA
UNIVERSITÀ DI ROMA

Task Singular Vectors

Reducing Task Interference in Model Merging

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Motivation and Aim

The Problem I Tackled

- Training separate models for each task is time consuming
- I wanted to combine multiple fine tuned models into one multi task model

The Challenge: Task Interference

- Simple averaging (Task Arithmetic) causes task updates to conflict in weight space
- This interference hurts overall performance

My Approach: Task Singular Vectors (TSV)

- TSV Compress: Reduces storage by 10× using SVD compression
- TSV Merge: Decorrelates task subspaces to reduce interference

Method Overview

How TSV Works (Paper: Gargiulo et al. 2024)

TSV Compress (Storage Efficiency)

- Applied truncated SVD to each task vector: $\Delta = U \cdot \Sigma \cdot V^T$
- Kept only top 10% of singular components
- 90% of data removed but I kept almost all important information

TSV Merge (Reducing Interference)

- I Computed SVD for each task's updates
- Stacked all task bases together
- Applied Procrustes whitening to make them orthogonal
- Conflicting tasks don't cancel each other out

Main Idea: By decorrelating subspaces I prevented task interference during merging

Implementation and Experiments

Component	What I Used
Model	CLIP ViT B/16 (86M parameters)
Datasets	5 vision tasks: MNIST, EuroSAT, DTD, GTSRB, SVHN
Training	5,000 samples per task for 2 epochs
Evaluation	1,024 samples per task
Baselines	Task Arithmetic and TIES Merging
Hardware	MacBook M4

Note: Smaller samples were chosen due to hardware limitation

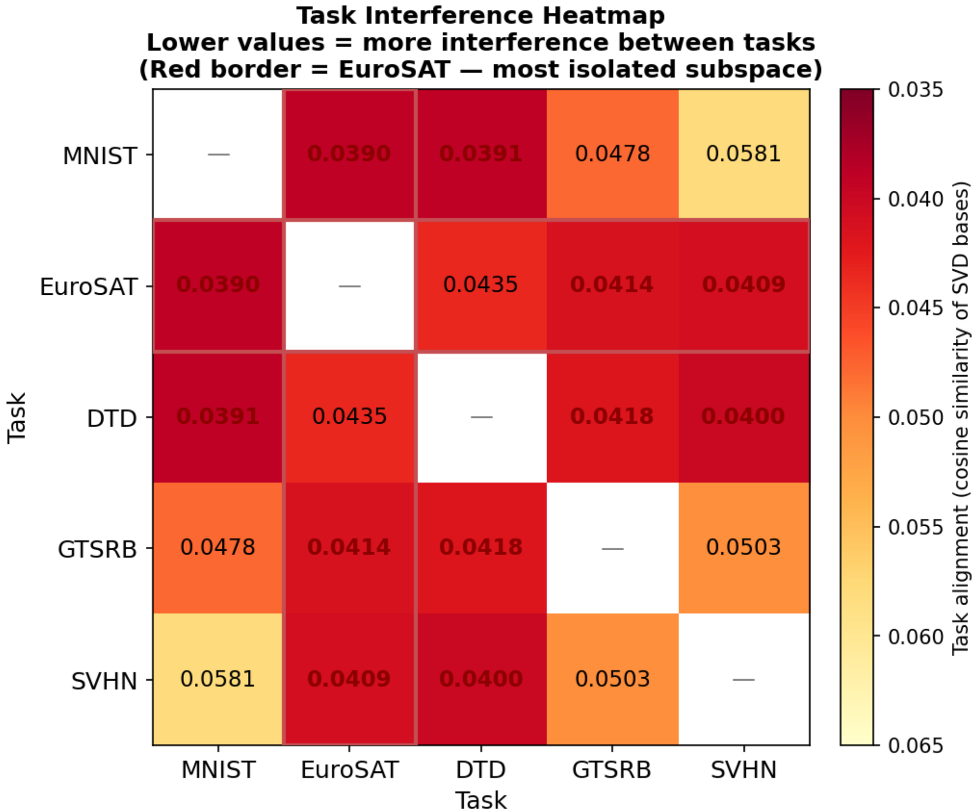
Results: My Key Finding

Method	Avg	MNIST	EuroSAT	DTD	GTSRB	SVHN
Zero shot	48.7%	52.3	51.3	44.9	42.7	52.2
Task Arithmetic	80.7%	97.4	73.9	58.8	86.8	86.8
TIES Merging	71.9%	93.8	68.8	51.2	70.2	75.7
My TSV Merge	**80.5%**	**95.9**	**91.6**	**58.4**	**77.8**	**78.5**

Headline Result: 91.6% on EuroSAT with TSV Merge in comparison with 73.9% with Task Arithmetic
That's a 17.7 percentage point improvement.

Why: Satellite imagery has unique features that conflict with other visual tasks.
TSV decorrelation specifically fixed this interference.

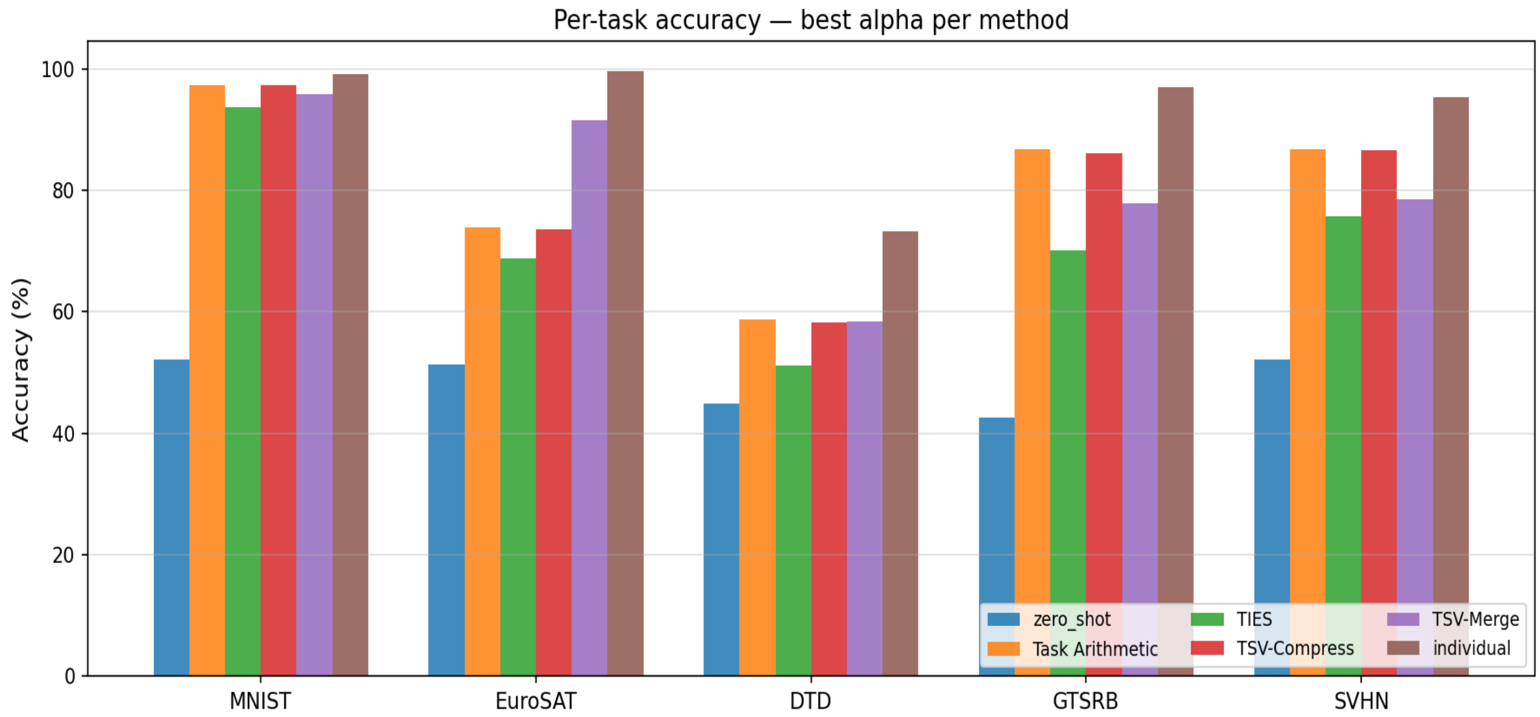
Visual Results: Task Interference Heatmap



Lower values = more interference between tasks. EuroSAT (red border) has the lowest alignment with all other tasks.

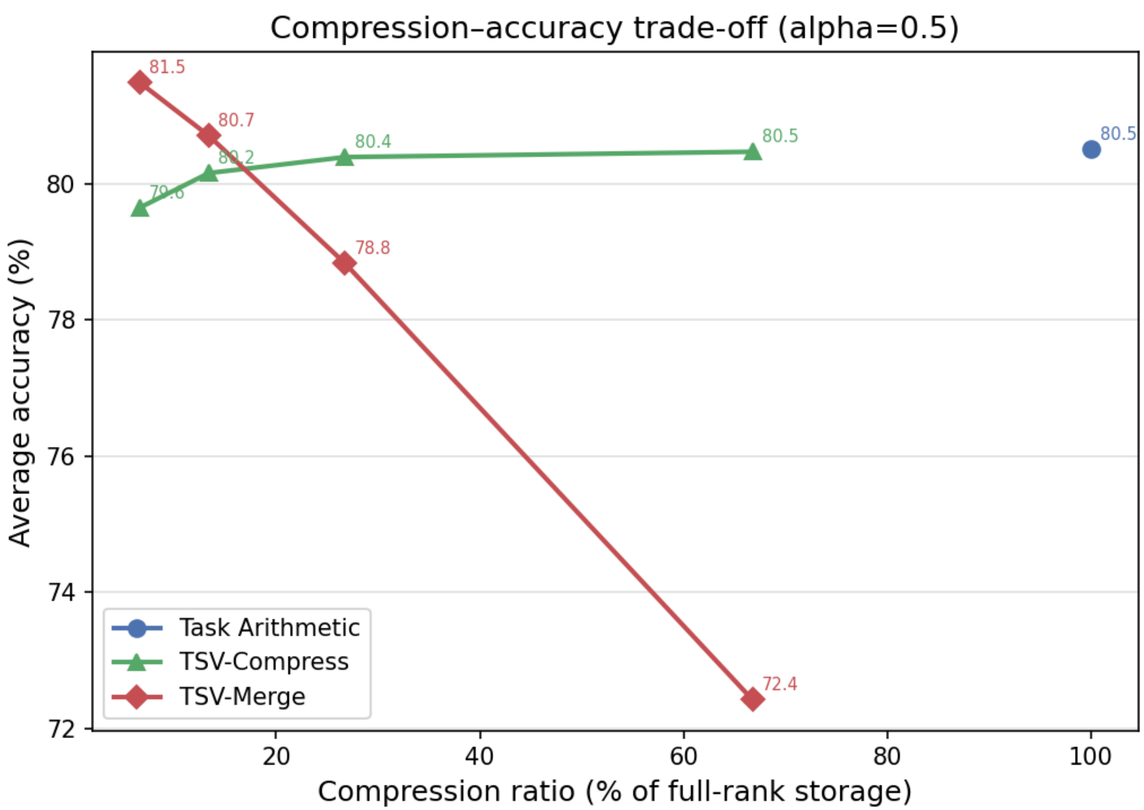
TSV Merge’s shows, whitening directly targets these misaligned subspaces explaining the 17.7pp gain on EuroSAT.

Visual Results: Per Task Breakdown



EuroSAT shows good improvement. MNIST, GTSRB, SVHN: Task Arithmetic works fine. DTD: hard for everyone. TSV Merge helps specifically when tasks have high interference.

Compression: 10× Storage Reduction



Key Findings:

TSV Merge at just 5% rank (6.6% of the original storage), it actually beats full Task Arithmetic with 81.5%.

TSV Compress is also similar, staying close to Task Arithmetic even at very low ranks

Result: Task vectors are highly compressible. Most layers captured 80%+ energy in top 10% of singular values.

Conclusions

What I Achieved

- Built TSV from scratch
- Verified compression works: 6.6% storage with minimal accuracy loss
- Demonstrated interference reduction: 17.7pp improvement on EuroSAT
- Compared against two strong baselines

When to Use TSV (My Recommendation)

- Use TSV Merge: High interference tasks from different domains
- Use TSV Compress: Storage or bandwidth limitations
- Use Task Arithmetic: Similar tasks, simpler approach

Limitations

- 5,000 training samples per task vs full datasets
- 1,024 test samples for speed
- 5 datasets instead of 8

THE-END